

# Data & Databases Final Project

## Methodology

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[Project link](#)  
[GitHub repository](#)

## Introduction

Who do you think the largest private landowners are in New York City? One can imagine the lists are filled with powerful [family real estate empires](#); perhaps rivaled only by the dominant companies they've cultivated. Maybe it's [private equity buying up residential properties](#) and fueling the ever-increasing NYC rent prices.

The reality? Columbia University, one of the city's oldest universities, claimed the top spot on the list as a result of large investments in a new campus extension further north into Harlem. NYU, another storied institution, has garnered a massive land holdings portfolio (worth an estimated [\\$8](#) - [\\$15](#) billion, thanks largely to ever-increasing property values in Greenwich Village.

Across the city, universities have quietly amassed massive real estate portfolios. And, thanks to a property tax exemption for educational nonprofits in New York City, the local government misses out on potentially hundreds of millions of dollars annually.

This project brings together data from multiple sources into one comprehensive, intuitive dashboard that allows non-technical viewers to explore university land holdings and discover the data for themselves.

## Sources

The data for this project comes from multiple public sources. None of the data is itself private, though each of the pieces is found through a separate, somewhat tedious search process, and no site aggregates it all together in this way.

The major sources are:

### Department of Housing Preservation and Development Data:

- [Multiple Dwelling Registrations](#)
- [Registration Contacts](#)

### **Automated City Register Information System Data:**

- [ACRIS Real Property Legals](#)
- [ACRIS Real Property Master](#)
- [ACRIS Real Property Parties](#)

The Department of Housing Preservation and Development keeps and makes public records about the owners of residential buildings that meet certain criteria. In the department's own words:

*"Pursuant to New York City's Housing Maintenance Code, the Department of Housing Preservation and Development (HPD) collects registration information from owners of residential rental units. Owners are required to register if they own residential buildings with three or more units or if they own one- or two-family homes and neither they nor members of their immediate family live there. Registrations are required upon taking ownership of a qualifying building, and once a year thereafter."*

This data contains information about any registered buildings, including their address, borough/block/lot numbers, registration ID and building ID. The other HPD database includes name and limited contact information about the owners who registered their buildings.

The other data sets – from the Automated City Register Information System – contains data about legal documents filed related to properties in the city. The Legals dataset is helpful for matching HPD data (based on Borough/Block/Lot) to document IDs. From there, the other two datasets have information about documents and related parties.

### **Data Gathering**

The data gathering phase started at the HPD data. Before downloading the datasets, the Registration Contacts data was filtered based on the "Corporation Name" column. In particular, I returned only results where the corporation name matched one of these results:

#### ***Columbia University:***

*"COLUMBIA UNIVERSITY"*

*"TEACHERS COLLEGE-COLUMBIA UNIV"*

*"THE TRUSTEES OF COLUMBIA UNIVERSITY"*

*"TRUSTEE OF COLUMBIA UNIVERSITY"*

*"TRUSTEES OF COLUMBIA UNIVERSITY"*

*"TRUSTEES OF COLUMBIA UNIVERSITY"*

*"TRUSTEES OF COLUMBIA UNIVERSITY CIT"*

***New York University:***

*“NYU REAL ESTATE CORP”*

*“NYU REAL ESATE CORP”*

*“NYU School of Law Foundation”*

*“NYU SCHOOL OF LAW FOUNDATION”*

*“New York University”*

*“NEW YORK UNIVESITY”*

***St. Johns University:***

*“St. Johns University”*

*“ST.JOHNS UNIVERSITY”*

*“ST JOHNS UNIVERSITY, NEW YORK”*

***Pratt Institute:***

*“PRATT INSTITUTE”*

*“PRATT INSTITUTE FACILITIES DEPARTMENT”*

***The New School:***

*“THE NEW SCHOOL”*

***Fordham University:***

*“FORDHAM UNIVERSITY”*

I then matched the exported Registration Contacts data with the full Multiple Dwelling Registrations data, based on the RegistrationID column.

For matching the ACRIS data, I combined the combined HPD data with the ACRIS Legals data set based on the borough, block and lot numbers of the buildings, then brought in the other two ACRIS data sets based on Document ID.

There are often multiple documents per building, so my ultimate dashboard shows information aggregated at the building level, with the option to view more about each building’s particular historical documents.

## **Scope: Population, Timeframe**

The current iteration of my dashboard shows 238 buildings across the 6 schools I’m focusing on. This is certainly an under-count (which one can observe by simply looking at the missing parcels on the map view of the dashboard), but I can be pretty certain that the buildings within my database do belong to the universities.

## **Critique**

Because I started my search with the HPD data, the universe of data I'm working in is primarily housing-related. While this is interesting (and universities obviously operate a lot of housing for their students), it might not totally reflect a university's land holdings.

I am in the process of reversing my search process: starting with ACRIS parties data and filtering that based on names that are associated with universities, then matching in other ACRIS data based on document ID and bringing in HPD or other data based on borough/block/lot information. This search procedure would cast a wider net (starting beyond housing-specific data), even though some of the non-residential buildings might not benefit from the additional contact information available from the HPD data sets.

The current process also requires human involvement to run. Ideally, this process would be automated and scheduled to run periodically on its own so as to keep the information in the dashboard fully up-to-date.

Finally, this procedure will successfully identify buildings that are openly listed in public data as owned by any of these universities. While there has not been widespread reporting or concern about NYC schools buying or owning land under aliases or shell LLCs, my search would not find any such properties. A work around could potentially be found (eg. by connecting related parties' names or by using other university resources like campus maps), but that would require further research and coding to become part of this project.

## **Dashboard Design**

The design of the dashboard is intended to make it easy for a user to interface with the data. The landing page for the dashboard is a map view of the data, along with an abbreviated stats display. A viewer can see building-specific information by interacting with the map itself, or else they can navigate to a view that displays all the buildings or all the entities as tables.

Opening with a map presents this data in the most intuitive way – geographically – while allowing for more thorough searching or filtering in a traditional table/sheet environment.

## **Conclusion**

This dashboard does, I think, meaningfully help non-technical users understand some of the scope of university land holdings in the city. While some more work is needed to expand my dataset beyond the HPD-first search I started with, even this look is helpful for readers. Whether it's the scores of blue dots near Columbia's Morningside Heights campus or the assortment of purple ones circling NYU's Greenwich Village sprawl, this tool shows some of the sheer amount of buildings owned by universities in the city, and meaningfully brings together historical records and legal documents in one intuitive, easy to use dashboard.